

Cloud Computing

Computing power and storage delivered over the internet

066

Cloud Computing delivers computing services — servers, storage, databases, and software — over the internet, allowing organizations to access resources on demand without owning physical infrastructure.

● DEFINITION

The delivery of computing services, including servers, storage, databases, and software, over the internet on a pay-as-you-go basis.

● WHY IT MATTERS

Cloud computing eliminates the need for organizations to build and maintain expensive physical data centers, dramatically lowering the barrier to deploying scalable technology.

● WORKING PRINCIPLE



Cloud providers operate massive data centers and rent out computing resources — processing power, storage, and software — on demand, allowing customers to scale usage up or down as needed and pay only for what they use.

● QUICK FACTS

- ▶ A significant share of the internet's traffic flows through just a few major cloud providers
- ▶ Cloud computing enabled many startups to launch globally without ever owning a physical server

● KEY TERMS

IaaS SaaS Scalability Data center

● CORE CONCEPTS

- ▶ Infrastructure as a Service (IaaS)
- ▶ Platform as a Service (PaaS)
- ▶ Software as a Service (SaaS)
- ▶ Scalability and elasticity

● APPLICATIONS

- ▶ Website and application hosting
- ▶ Data storage and backup
- ▶ Big data processing
- ▶ Software-as-a-service platforms

● ADVANTAGES

- ▶ Eliminates need for costly physical infrastructure
- ▶ Scales instantly with demand
- ▶ Enables global access and collaboration

CAREER OPPORTUNITIES

Cloud engineer, Cloud architect, DevOps engineer, Cloud security specialist

RESEARCH AREAS

Edge computing, Multi-cloud architecture, Cloud cost optimization

● REAL-LIFE EXAMPLES

- ▶ Streaming services like Netflix running on cloud infrastructure
- ▶ Cloud file storage services
- ▶ Business software delivered via subscription (SaaS)

● LIMITATIONS

- ▶ Ongoing subscription costs can add up over time
- ▶ Requires reliable internet connectivity
- ▶ Raises data security and privacy considerations

INDUSTRY APPLICATIONS

Every digital-first industry, Startups, Enterprise IT, Government

FUTURE SCOPE

Continued migration of enterprise workloads to the cloud, alongside growth of edge computing for latency-sensitive applications.

● CURRENT TRENDS

Rapid growth of AI workloads running on specialized cloud infrastructure designed for machine learning training and inference.

● SUMMARY

Cloud computing delivers computing power and storage over the internet on demand, letting organizations scale technology without owning physical infrastructure.